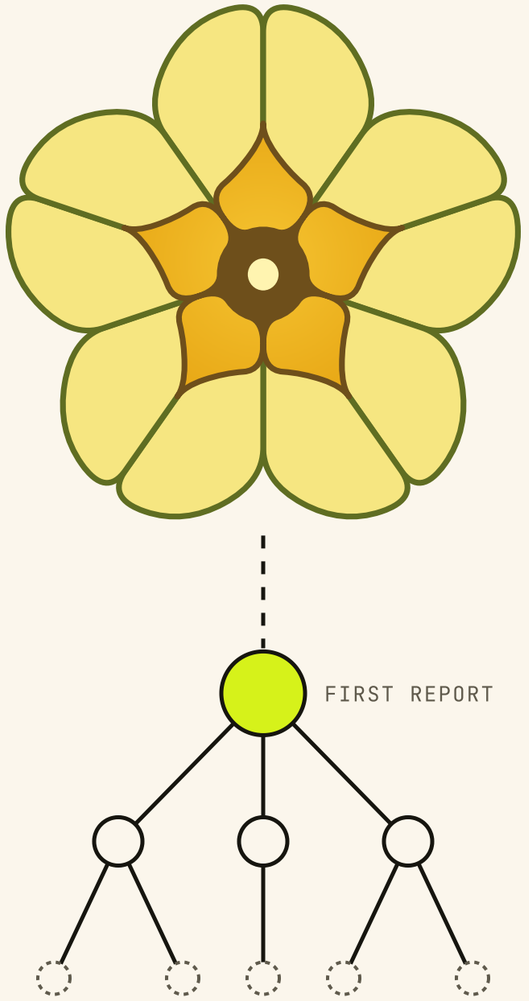
Who said it *first?*

Tracing a statement back to the article that reported it first, with a loop that makes the tracer better on its own.



PREPARED BY **Dr. Aleks Jakulin** data.Flowers

WITH **Kevin Chiu** Plastachio

BUILT AT AN RSIHOUSE × SUNDAI CLUB HACK

We trace a statement back to the article that reported it *first*

“X SAID THAT…” ---- SHORTLIST 30 ---- READ ---- FIRST REPORT *website · date · time*

THE PROBLEM

Our model of reality is built from retold text. Quotes get trimmed, reworded and credited to the wrong person, and AI models learn the drifted version.

WHAT WE BUILT

A tracer over four months of news: about 4 million articles, January to April 2020. Give it “X said that…”; it finds the quotes that say it and the earliest article that carried them.

HOW IT IMPROVES

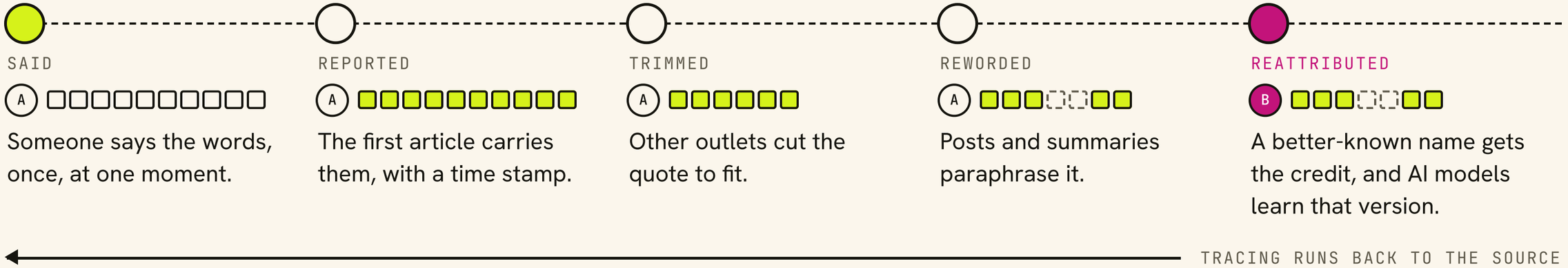
A stronger AI model studies the tracer's mistakes and suggests small fixes. Three fixes passed, and on questions it never studied, correct first reports rose from 60% to 85%.

WHERE IT LEADS

News is the simple case. Next: the origins of ideas and inventions, as long as they are shared in the open.

Our model of reality drifts, one retold quote at a *time*

HOW A QUOTE DRIFTS



THE PROBLEM

How do we keep our model of reality aligned with reality itself?

THE SUBPROBLEM

Identify a quote, track it through time, and find its source.

WHY NOW

Kevin Chiu gave eleven AI models from different labs the same simple puzzle. All of them chose the same wrong answer, most likely because they learned from the same text.

MemeTracker showed how quotes travel; Quotebank tells us who *said* them

INSPIRATION · MEMETRACKER, 2009

Quotes travel in families, and the shortest version travels furthest

11,235	“lipstick on a pig”
1,051	“you can put lipstick on a pig”
967	“you can put lipstick on a pig but it's still a pig”
604	“i think they put some lipstick on a pig but it's still a pig”
542	“putting lipstick on a pig”

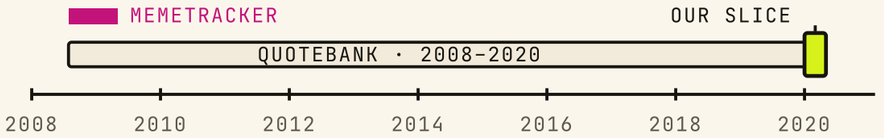
5 of 61 versions in one family, with mentions. The study grouped 90 million articles from 1.6 million news sites and blogs into such families. News outlets peaked about 2.5 hours before blogs.

DATA · QUOTEBANK, 2021

178 million quotes, each with its likely speaker

178M	162M	2008–20	377k
unique quotes	news articles	twelve years	websites

Speakers are linked to Wikidata. In the authors' test, 86.9% of quotes went to the right speaker. Open data (CC BY 4.0); the public release is larger still, at 235M quotes.



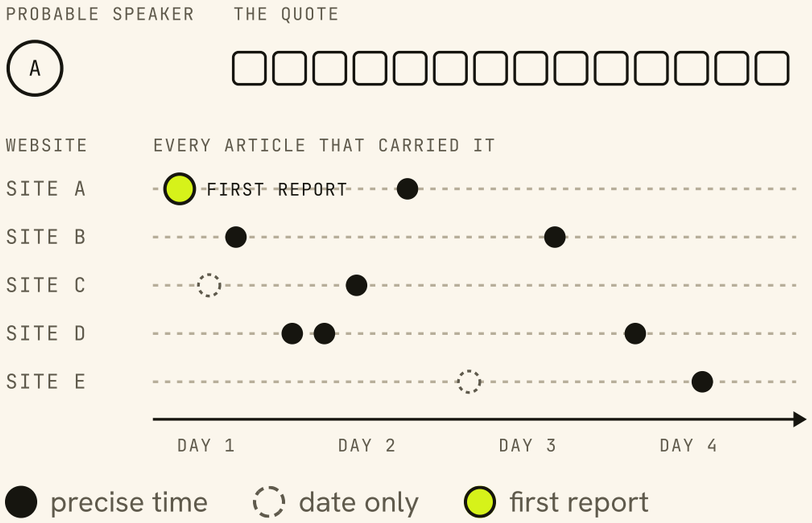
January–April 2020: about 4 million articles, small enough to iterate on in a day.

For every quote we know who probably said it, where it ran and exactly *when*

THE LIBRARY · JANUARY–APRIL 2020

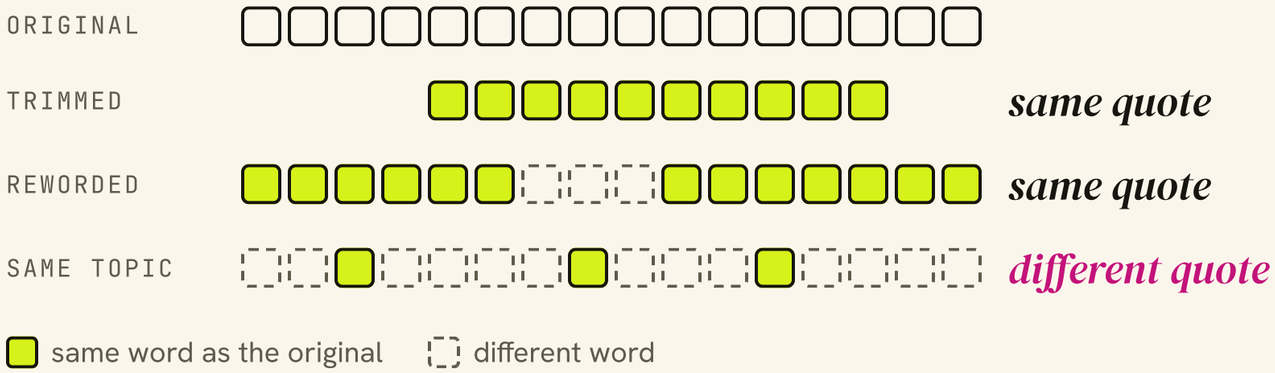
≈4M news articles

5.2M QUOTES · 16.7M MENTIONS WITH TIME, OUTLET, URL



GROUPING VERSIONS OF THE SAME QUOTE

Outlets trim quotes or reword them slightly. If two quotes share **long runs of identical words** (eight or more in a row), we treat them as versions of one quote, so a shortened version still counts.



Three steps take a statement to its first *report*

1

SHORTLIST

Find 30 candidates



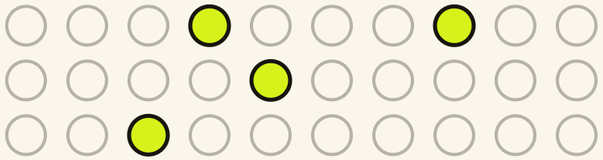
Given a sentence like “X said that…”, pull the 30 quotes that share the most distinctive words with it.

Best clue: the speaker's name.
Only quotes that existed before the statement was published.

2

CAREFUL READING

Keep the ones that say it



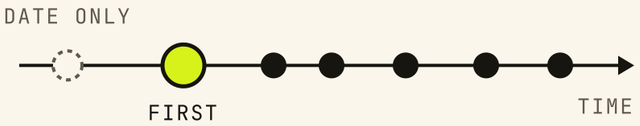
A small, fast AI model reads the shortlist and picks the quotes that actually say what the statement claims.

The same claim, not just the same topic.

3

FIRST REPORT

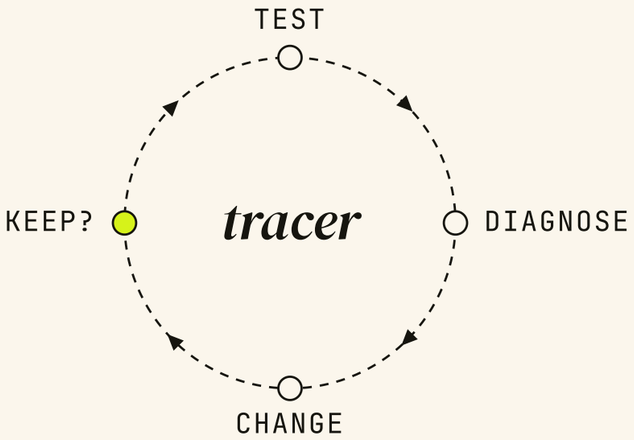
Pick the earliest article



For the quotes it picked, list every article that carried them and choose the earliest one.

Only articles with a precise publication time.

Only the reading step uses an AI model. The shortlist is a word search, the first report a sort by time.



The tracer improves itself, and a locked test keeps it *honest*

TEST	364 statements where we already know which article came first, split into three sets.
DIAGNOSE	A stronger AI model looks at the mistakes and works out their common cause.
CHANGE	It suggests a small change to how the shortlist is built or how the checker reads.
KEEP?	Only if it fixes clearly more questions than it breaks (beyond run-to-run noise), and does no worse on the second set.

PRACTICE SET · 145

The stronger model studies mistakes here.

START 58%

NOW 83%

SECOND SET · 73

Never studied; a change that does worse here is dropped.

START 60%

NOW 85%

FINAL SET · 146 · LOCKED

Scored once at the start, once at the very end.

START 71%

END SCORED AT THE VERY END

The gains held on statements the loop never *saw*



60% → 85%
Watch set: 100 statements, never read by the loop

60% → 85%
Second set: 73 statements, never studied

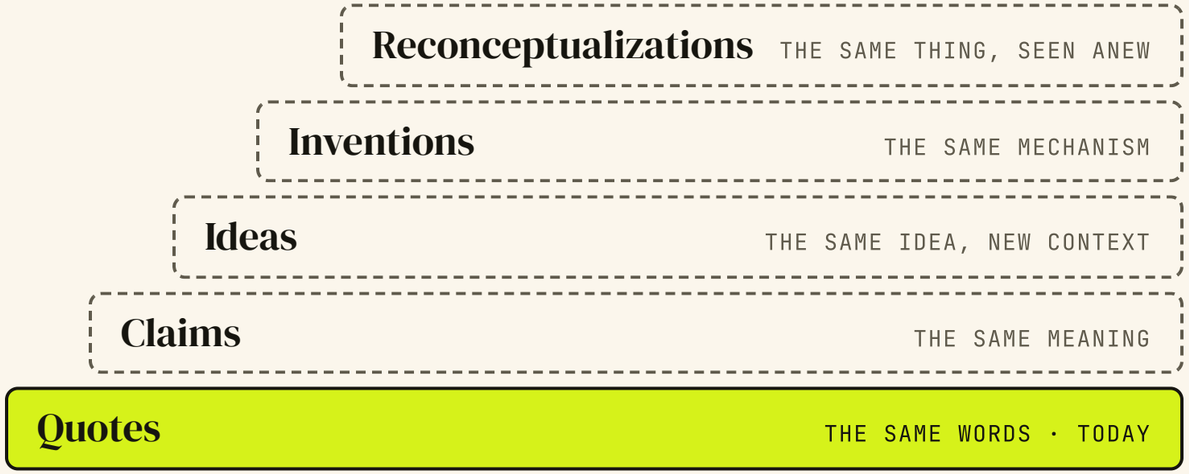
58% → 83%
Practice set: 145 statements

3 kept · 16 dropped
Changes the stronger model proposed

Three lines ran in parallel. Tuning everything at once (blue) beat tuning only the shortlist (orange, 74%) or only the checker (green, 68%).

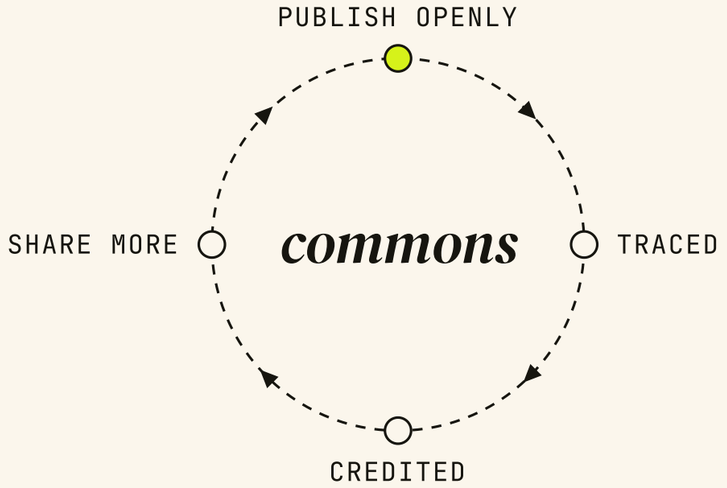
News is the simple case; next we can trace where ideas and inventions *come from*

FROM WORDS TO IDEAS



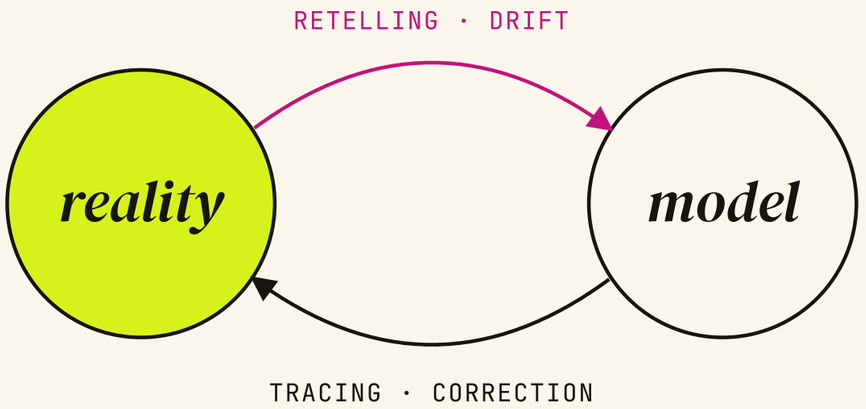
Each step up needs a looser sense of “the same”. The method stays: find the versions, order them in time, check the earliest.

ONLY IN THE OPEN



We can only trace what is published openly. Then credit goes to whoever shared an idea first, and that rewards contributing to the commons.

The same loop can keep any model of reality *aligned*



NEXT

1. Score the locked final set.
2. Spot wire and press-release copies by their shared text.
3. Map which outlets copy which.
4. Follow paraphrased facts, not only quotes.
5. Add the other twelve years of Quotebank, 2008–2019.

NEWSROOMS

Check who said it, and where it first ran, before repeating it.

AI ANSWERS

Cite the first report, not the tenth copy.

TRAINING DATA

See which claims many models learned from the same copy.

PUBLIC DATA

data.Flowers runs the same loop so official data stays the source.

THE IDEA **A claim traced to its first report is a claim our models can't quietly drift on.**